6 properties
Mechanical
Electrical
Thermal
optical
deterioration

material selection
- categorization & service
- cost effectiveness
- deterioration

advanced materials
- nano
- bio
- smart
- semiconductors

materials
- metals
- ceramics
- polymers
- composites

MTL - 200
MATERIALS SCIENCE FUNDAMENTALS
TERM TEST NO. 1

Professor C. Ravindran

Duration: 45 mts.

structure
- subatomic
- atomic
- microscopic
- macroscopic

All questions carry equal marks.

1. Calculate the radius of a tantalum atom, given that Ta has a BBC crystal structure, a density of 16.6 g/cm³, and an atomic weight of 180.9 g/mol.

2. Some metal is known to have a cubic unit cell with an edge length of 0.3524 nm. Furthermore, it has a density of 8.90 g/cm³ and an atomic weight of 58.71 g/mol. Which of the metals in the table shown below is this metal? Why?

TABLE 3.1 Atomic Radii and Crystal Structures for 16 Metals

Metal	*Crystal Structure*[a]	*Atomic Radius*[b] *(nm)*	*Metal*	*Crystal Structure*	*Atomic Radius (nm)*
Aluminum	FCC	0.1431	Molybdenum	BCC	0.1363
Cadmium	HCP	0.1490	Nickel	FCC	0.1246
Chromium	BCC	0.1249	Platinum	FCC	0.1387
Cobalt	HCP	0.1253	Silver	FCC	0.1445
Copper	FCC	0.1278	Tantalum	BCC	0.1430
Gold	FCC	0.1442	Titanium (α)	HCP	0.1445
Iron (α)	BCC	0.1241	Tungsten	BCC	0.1371
Lead	FCC	0.1750	Zinc	HCP	0.1332

[a] FCC = face-centered cubic; HCP = hexagonal close-packed; BCC = body-centered cubic.
[b] A nanometer (nm) equals 10^{-9} m; to convert from nanometers to angstrom units (Å), multiply the nanometer value by 10.

fcc
$a = 2\sqrt{2}R$
12
bcc
$a = \frac{4}{\sqrt{3}}r$
8
HCP
$a = 2R$
12

Kame house (ha)
" "
turtle lazer shout!

3. The metal rhodium has an FCC crystal structure. If the angle of diffraction for the (311) set of planes occurs at 36.12° (first-order reflection) when monochromatic x-radiation having a wavelength of 0.0711 nm is used, compute (a) the interplanar spacing for this set of planes, and (b) the atomic radius for a rhodium atom.

4. Nickel forms a substantial solid solution with copper. Compute the number of nickel atoms per cubic centimeter for a copper-nickel alloy that contains 1.0 wt% Ni and 99.0 wt% Cu. The densities of pure nickel and copper are 8.90 and 8.93 g/cm³, respectively. The atomic weights of ~~nickel and copper~~ are 63.54 and 58.68 g/mol respectively.

Copper and nickel

Bragg law $n\lambda = 2d_{hkl}\sin\theta$

2θ - diffraction angle

#2

$$a = 0.3524\,nm$$
$$V_c = a^3$$

$$\rho = \frac{nA}{V_c N_A}$$

$$\therefore n = \frac{\rho a^3 N_A}{A}$$

$$n = \frac{(8.90\,g/cm^3)(0.3524 \times 10^{-7}\,cm)^3 (6.023 \times 10^{23}\,atoms/mol)}{58.71\,g/mol}$$

$$= 4.0$$

Since the number of atoms/unit cell = 4.0, the crystal structure is FCC.

$$a^2 + a^2 = (4R)^2$$
$$a = 2R\sqrt{2}$$

Atomic Radius, $R = \frac{a}{2\sqrt{2}} = \frac{0.3524}{2\sqrt{2}} = 0.1246\,nm$

This matches the atomic radius for Nickel in the tables, which also has an FCC structure.

#4 99 % Cu 1 % Ni

99 gm Cu 1 gm Ni

Atomic Wts: 63.54 g/mol Cu 58.69 g/mol Ni

$$N_m(Cu) = \frac{m}{A} = \frac{99\,gm}{63.54\,g/mol} = 1.56\,mol$$

$$N_m(Ni) = \frac{m}{A} = \frac{1\,gm}{58.69\,g/mol} = 1.704\times10^{-2}\,mol$$

$$\text{Atom Fraction of Ni} = \frac{1.704\times10^{-2}\,mol}{1.704\times10^{-2} + 1.56}$$

$$= 1.081\times10^{-2}\ \text{Ni atoms/Cu atom}$$

This is just the fraction of Cu atoms which are replaced with Ni atoms in solid solution. The product of this atom fraction and the number of copper atoms per cm^3 (N_{Cu}) yield the number of Ni atoms per cm^3 (N_{Ni}) for this alloy.

$$N_{Cu} = \frac{\rho_{Cu} N_A}{A_{Cu}}$$

$$N_{Cu} = \frac{(8.93\,g/cm^3)(6.023\times10^{23}\,atoms/mol)}{63.54\,g/mol}$$

$$= 8.46\times10^{22}\ \text{Cu atoms}/cm^3$$

No. of Ni atoms = Ni atom fraction × N_{Cu}

$$N_{Ni} = (1.081\times10^{-2}\ \text{Ni atoms/Cu atom})(8.46\times10^{22}\ \text{Cu atoms}/cm^3)$$

$$= 9.15\times10^{20}\ \text{Ni atoms}/cm^3$$